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DEPARTMENT OF
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Course Project Report

"Credit Card Fraud Detection and Transaction Analysis System"

Project Associates:

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03	U Anjali	3BR24AI168
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Under the Guidance of

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Department of AI&ML
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Visvesvaraya Technological University

Belagavi, Karnataka

2025-2026

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Department of Artificial Intelligence & Machine Learning

Academic year 2025-26

Rubrics for Course Project

Course: Introduction to Data Analytics **Course Code:**22AI463

Batch No. : 2

Project Title
Credit Card Fraud Detection and Transaction Analysis System

USN	Name	Problem Identification & Objective Formulation **(5)**	Dataset Collection & Preprocessing **(5)**	Application of Data Analytics Techniques **(7)**	Visualization & Interpretation of Results **(4)**	Project Presentation, Report & Viva **(4)**	Total Marks **(25)**
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Signature of the Faculty

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

CERTIFICATE

This is to certify that the Course Project entitled "**Credit Card Fraud Detection and Transaction Analysis System**" has been successfully completed for the course **Introduction to Data Analytics** by **Sandhya R, Tanzeem Tabassum, U Anjali, Ulfath Zaara, Vanishree,** bearing **3BR24AI141, 3BR24AI166, 3BR24AI168, 3BR24AI172, 3BR24AI177,** students of IV semester B.E. for the partial fulfillment of the requirements for the Bachelor of Engineering in Artificial Intelligence & Machine Learning during the academic year 2025-26.

Mr K R Thukaram Rao	**Dr. Vidyavathi**
Course Coordinator	Head of the Department

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ABSTRACT

Credit card fraud has become a major issue in digital banking and online payment systems. The main purpose of this project is to analyze credit card transaction data and identify fraudulent transaction patterns using data analytics techniques. The methodology involves collecting transaction datasets, performing data cleaning and preprocessing, and analyzing the data using Power BI. Various visualizations such as bar charts, pie charts, line graphs, and heat maps are used to study fraud trends, transaction behavior, and high-risk activities. The findings of the project help and understanding the difference between genuine and fraudulent transactions based on transaction amount, frequency, and time patterns. The dashboard provides clear insights into fraud distribution and suspicious activities through interactive reports and analytical views. The conclusion of the project shows that data analytics and business intelligence techniques can effectively support fraud analysis and help financial organizations improve transaction monitoring and decision-making processes. The system is simple, efficient, and suitable for real-world financial analytics applications.

PROBLEM STATEMENT

To Design and Develop a Credit Card Fraud Detection System that analyzes transaction data to identify fraudulent activities and provide actionable insights through interactive dashboards and data visualization.

1. INTRODUCTION

The rapid growth of digital payment systems and online banking has significantly increased the use of credit cards for financial transactions. While credit cards provide convenience and flexibility, they have also become a target for fraudulent activities. Credit card fraud results in financial losses for customers and organizations, making fraud detection an important aspect of financial security.

This project focuses on analyzing credit card transaction data using Power BI to identify fraudulent and non-fraudulent transactions. The dataset was collected from Kaggle and processed using Power Query for data cleaning and transformation. Various measures such as total transactions, fraud count, non-fraud count, and fraud percentage were calculated and visualized through an interactive dashboard.

In today's data-driven environment, visual analytics has become an essential tool for detecting unusual transaction patterns and reducing financial risks. By presenting transaction data in a structured and interactive format, organizations can quickly identify potential fraud cases and take appropriate actions. The use of Power BI enhances data interpretation and enables users to gain deeper insights through real-time dashboards and reports.

The dashboard includes KPI cards, charts, and tables that provide a clear overview of transaction patterns and fraud distribution. These visualizations help users quickly identify suspicious transactions and understand the overall transaction behavior.

The project demonstrates how business intelligence and data visualization techniques can be used to monitor financial transactions effectively. By converting raw data into meaningful insights, Power BI helps organizations improve fraud monitoring, reporting, and decision-making processes.

2. OBJECTIVES

1. To collect and analyze the Credit Card Fraud Detection dataset.
2. To clean and preprocess the transaction data using Power Query.
3. To identify and classify fraudulent and non-fraudulent transactions.
4. To develop an interactive Power BI dashboard for effective data visualization.
5. To generate meaningful insights that support fraud monitoring and decision-making.

3. DATASET COLLECTION

The Credit Card Fraud Detection dataset was collected from Kaggle, a widely used platform for data science and analytics projects. The original dataset contained a large number of transaction records categorized as fraudulent and non-fraudulent transactions. Before beginning the analysis, the dataset was carefully studied to understand its structure, attributes, and fraud distribution.

To make the dataset suitable for dashboard development and analysis, a subset of the original dataset was extracted. While selecting the records, the ratio between fraudulent and normal transactions was maintained to ensure that both categories were properly represented in the analysis. Special attention was given to include all available types of fraudulent transactions present in the selected data so that different fraud scenarios could be analyzed effectively.

After extracting the required records, the dataset was organized and saved in CSV format for further processing in Microsoft Power BI. The extracted dataset consisted of both fraud and non-fraud transactions, enabling meaningful comparison and visualization of transaction behavior. This prepared dataset served as the foundation for data preprocessing, analysis, and dashboard creation.

Dataset Extraction Process:

1. Downloaded the Credit Card Fraud Detection dataset from Kaggle.
2. Studied the dataset structure and transaction attributes.
3. Selected a representative subset of transaction records.
4. Maintained the proportion of fraud and normal transactions in the extracted data.
5. Included all available fraud transaction categories present in the selected dataset.
6. Saved the extracted data in CSV format.
7. Imported the dataset into Power BI for further processing and analysis.

Time	V1	V2	V3	V4	V10	V12	V14	Amount	Class
0	-1.3598071336738	-0.0727811733098497	2.53634673796914	1.37815522427443	0.0907941719789316	-0.617800855762348	-0.311169353699879	149.62	0
1	1.19185711131486	0.26615071205963	0.16648011335321	0.448154078460911	-0.166974414004614	1.06523531137287	-0.143772296441519	2.69	0
2	-1.35835406159823	-1.34016307473609	1.77320934263119	0.379779593034328	0.207642865216696	0.066083685268831	-0.165945922763554	378.66	0
3	-0.966271711572087	-0.185226008082898	1.79299333957872	-0.863291275036453	-0.0549519224713749	0.178228225877303	-0.28792374549456	123.5	0
4	-1.15823309349523	0.877736754848451	1.548717846511	0.403033933955121	0.753074431976354	0.53819555014995	-1.11966983471731	69.99	0
5	-0.425965884412454	0.960523044882985	1.14110934232219	-0.168252079760302	-0.371407196834471	0.359893837038039	-0.137133700217612	3.67	0
6	1.22965763450793	0.141003507049326	0.0453707735899449	1.20261273673594	-0.0992543211289237	-0.153825826253651	0.16737196252175	4.99	0
7	-0.644269442348146	1.41796354547385	1.0743803763556	-0.492199018495015	1.24937617815176	0.291474353088705	-1.32386521970526	40.8	0
8	-0.89428608220282	0.286157196276544	-0.113192212729871	-0.271526130088604	-0.410430432848439	-0.110452261733098	0.0743553603016731	93.2	0
9	-0.33826175242575	1.11959337641566	1.04436655157316	-0.222187276738296	-0.366845639206541	0.836389570307029	-0.443522816876142	3.68	0
10	1.44904378114715	-1.17633882535966	0.913859832832795	-1.37566665499943	1.62665905834133	-0.671439778462005	-0.0950450453999549	7.8	0
11	0.38497821518095	0.616109459176472	-0.874299702595052	-0.0940186259679115	0.30975539423728	-0.326143233995877	0.362832368569793	9.99	0
12	1.249998742053	-1.22163680921816	0.383930151282291	-1.23489868766892	1.32372927445937	-0.242681998944186	-0.317630527025074	121.5	0
13	1.0693735878819	0.287722129331455	0.828612726634281	2.71252042961718	0.460230444301678	0.32338724546722	-0.178485175177916	27.5	0
14	-2.7918547659339	-0.327770756658658	1.64175016056605	1.76747274389883	1.1510869876677	0.7929439518176	-0.734975105820311	58.8	0
15	-0.752417042956605	0.345485415344747	2.05732291276727	-1.46864329840046	0.747730827192802	-0.770406728847129	-1.06660368148653	15.99	0
16	1.10321543528383	-0.0402962145973447	1.2673320885949	1.28909146962552	-0.267975066537173	0.936707714991982	-0.468647287707221	12.99	0
17	-0.436905071360625	0.918966212909322	0.92459077438817	-0.727219053596792	-0.737979823596458	0.277192107214981	-0.291896460370468	0.89	0
18	-5.40125766315825	-5.45014783420644	1.18630463143652	1.73623880012095	0.345172827050629	0.970116716069048	-0.479129929276704	46.8	0
19	1.4929359769862	-1.02934573189487	0.45479473374366	-1.43802587991702	1.63807603690446	-0.63204651464934	0.0520105153724404	5	0
20	0.694884775607337	-1.36181910308009	1.02922103956032	0.834159299216716	0.568520735086962	1.29832870056251	-0.372650997239682	231.71	0
21	0.962496069914852	0.32846102605212	-0.17147905415064	2.10920406774016	0.724396314588903	0.406773575635315	0.98373941913199	34.09	0
22	-2.3122265423263	1.95199201064158	-1.60985073229769	3.9979055875468	-2.77227214465915	-2.89990738849473	-4.28925378244217	0	1
23	1.16661638244228	0.502120087854101	-0.0673003143663533	2.26156923949128	0.922174967079328	-0.531377250104474	1.12687010488156	2.28	0
24	0.247491127783665	0.277665627353681	1.18547084217971	-0.0926025498576041	-0.82988060074647	0.524933232159904	0.0813930875646215	22.75	0
25	-1.94652513121534	-0.0449005054418194	-0.405570068378956	-1.01305733702394	0.573742508037695	-0.215745003282808	0.0338977566837455	0.89	0
26	-2.0742946722629	-0.121481799450951	1.32202063048967	0.410007514171835	0.149450615348245	-0.180523156037298	-0.27979685563853	26.43	0

Table 3.1: Extracted data

4. SOFTWARE REQUIRED

- Microsoft Power BI Desktop – Data visualization and dashboard creation
- Microsoft Excel – Dataset storage and preprocessing
- Operating System: Microsoft Windows 10/11

5. IMPLEMENTATION

The implementation phase of the Credit Card Fraud Detection Dashboard involved collecting the dataset, performing data cleaning and transformation using Power Query, generating analytical measures, and creating interactive visualizations in Microsoft Power BI. The objective of this phase was to convert raw transaction data into meaningful insights that help identify fraudulent transactions and understand transaction behavior.

5.1 Preprocessing of the Data

Data preprocessing was performed using Power Query Editor in Power BI. Data cleaning and transformation are important steps because the quality of insights depends on the quality of data.

The following preprocessing operations were carried out:

Data Type Verification

The data types of all columns were checked and corrected.

- Time → Whole Number
- Fraud_Status → Whole Number
- V1, V2, V3, V4, V10, V12, V14 → Decimal Number
- Amount → Decimal Number

Column Renaming

The existing **Class** column was renamed to **Fraud_Status** to improve readability and make the dataset easier to understand.

Creating Transaction_Type Column

A new column named **Transaction_Type** was created.

The following logic was used:

- Fraud_Status = 1 → Fraud
- Fraud_Status = 0 → Normal

This helped categorize transactions in a more user-friendly manner.

Creating Index Column

An Index column was added starting from 1.

Benefits:

- Preserves original transaction order.
- Provides a unique identifier for each transaction.
- Simplifies referencing transaction records.

Applying Changes

After completing all transformation activities, the cleaned dataset was loaded into Power BI using the **Close & Apply** option.

Step	Transformation
1	Import Dataset
2	Verify Data Types
3	Rename Class Column
4	Create Transaction_Type
5	Add Index Column
6	Reposition Index Column
7	Apply Changes

Table 5.1: Power Query Transformation Steps

Time	V1	V2	V3	V4	V10	V12	V14	Amount	Fraud_Status	Transaction_Type	Index
0	-1.3598071336738	-0.0727811733098497	2.53634673796914	1.37815522427443	0.0907941719789316	-0.617800855762348	-0.311169353699879	149.62	0	Normal	1
1	1.19185711131486	0.26615071205963	0.16648011335321	0.448154078460911	-0.166974414004614	1.06523531137287	-0.143772296441519	2.69	0	Normal	2
2	-1.35835406159823	-1.34016307473609	1.77320934263119	0.379779593034328	0.207642865216696	0.066083685268831	-0.165945922763554	378.66	0	Normal	3
3	-0.966271711572087	-0.185226008082898	1.79299333957872	-0.863291275036453	-0.0549519224713749	0.178228225877303	-0.28792374549456	123.5	0	Normal	4
4	-1.15823309349523	0.877736754848451	1.548717846511	0.403033933955121	0.753074431976354	0.53819555014995	-1.11966983471731	69.99	0	Normal	5
5	-0.425965884412454	0.960523044882985	1.14110934232219	-0.168252079760302	-0.371407196834471	0.359893837038039	-0.137133700217612	3.67	0	Normal	6
6	1.22965763450793	0.141003507049326	0.0453707735899449	1.20261273673594	-0.0992543211289237	-0.153825826253651	0.16737196252175	4.99	0	Normal	7
7	-0.644269442348146	1.41796354547385	1.0743803763556	-0.492199018495015	1.24937617815176	0.291474353088705	-1.32386521970526	40.8	0	Normal	8
8	-0.89428608220282	0.286157196276544	-0.113192212729871	-0.271526130088604	-0.410430432848439	-0.110452261733098	0.0743553603016731	93.2	0	Normal	9
9	-0.33826175242575	1.11959337641566	1.04436655157316	-0.222187276738296	-0.366845639206541	0.836389570307029	-0.443522816876142	3.68	0	Normal	10
10	1.44904378114715	-1.17633882535966	0.913859832832795	-1.37566665499943	1.62665905834133	-0.671439778462005	-0.0950450453999549	7.8	0	Normal	11
11	0.38497821518095	0.616109459176472	-0.874299702595052	-0.0940186259679115	0.30975539423728	-0.326143233995877	0.362832368569793	9.99	0	Normal	12
12	1.249998742053	-1.22163680921816	0.383930151282291	-1.23489868766892	1.32372927445937	-0.242681998944186	-0.317630527025074	121.5	0	Normal	13
13	1.0693735878819	0.287722129331455	0.828612726634281	2.71252042961718	0.460230444301678	0.32338724546722	-0.178485175177916	27.5	0	Normal	14
14	-2.7918547659339	-0.327770756658658	1.64175016056605	1.76747274389883	1.1510869876677	0.7929439518176	-0.734975105820311	58.8	0	Normal	15

Table 5.2: Cleaned and transformed data

5.2 Processing Data and Generating Insight Calculations

After completing the data cleaning and transformation process, the dataset was processed in Power BI to generate meaningful insights. Various calculations and measures were created to analyze transaction data and understand the distribution of fraudulent and non-fraudulent transactions. These calculations helped summarize the data and provided a clear overview of transaction behavior.

Power BI's Data Analysis Expressions (DAX) were used to create measures that automatically calculate important metrics from the dataset. These measures formed the foundation for dashboard visualizations and reporting.

The following key calculations were performed:

1. **Total Transactions**

This measure calculates the total number of transactions available in the dataset. It provides an overall view of the transaction volume considered for analysis.

```
Total Transactions = COUNTROWS('dataset')
```

Result: 71 Transactions

2. **Fraud Count**

This measure calculates the total number of transactions identified as fraudulent based on the Fraud_Status column.

```
Fraud Count = COUNTROWS(FILTER('dataset','dataset'[Fraud_Status]=1))
```

Result: 3 Fraudulent Transactions

3. **Non-Fraud Count**

This measure calculates the number of legitimate or normal transactions present in the dataset.

```
Non Fraud Count = COUNTROWS(FILTER('dataset','dataset'[Fraud_Status]=0))
```

Result: 68 Non-Fraudulent Transactions

4. **Fraud Percentage**

This measure calculates the percentage of fraudulent transactions with respect to the total number of transactions.

```
Fraud percentage = DIVIDE([Fraud Count],[Total Transactions])*100
```

Result: 4.23%

Metric	Value
Total Transactions	71
Fraud Count	3
Non-Fraud Count	68
Fraud Percentage	4.23%

Table 5.3.: Generated Insights

The generated insights indicate that the majority of transactions in the dataset are legitimate, while only a small percentage are fraudulent. These calculations provide a quick summary of fraud distribution and help users understand the overall transaction pattern.

The calculated measures were further used in KPI cards, charts, and tables to create an interactive dashboard. These visualizations make it easier to monitor transaction activity, compare fraud and non-fraud records, and identify suspicious transactions effectively. The generated insights serve as the basis for decision-making and fraud analysis within the dashboard.

6. DASHBOARD VISUALIZATIONS

After processing the dataset and generating the required calculations, various visualizations were created in Microsoft Power BI to present fraud-related insights in a clear and interactive manner. Data visualization helps users understand transaction patterns, compare fraud and non-fraud activities, and identify suspicious transactions quickly.

The dashboard was designed using multiple visual components, each serving a specific analytical purpose. The visualizations transform raw transaction data into meaningful information that supports fraud analysis and decision-making.

1. **KPI Cards**

KPI Cards were used to display important metrics at the top of the dashboard. These cards provide a quick summary of the dataset and help users understand the overall transaction statistics.

The KPI Cards display:

- Total Transactions = 71
- Fraud Count = 3
- Non-Fraud Count = 68
- Fraud Percentage = 4.23%

Figure 6.1: KPI Cards Dashboard

7. Fraud vs Non-Fraud Transaction Bar Chart

A Bar Chart was created to compare fraudulent and non-fraudulent transactions. This visualization clearly shows the difference between the number of fraud and normal transactions in the dataset.

The chart helps users quickly identify the proportion of fraud cases and understand transaction distribution.

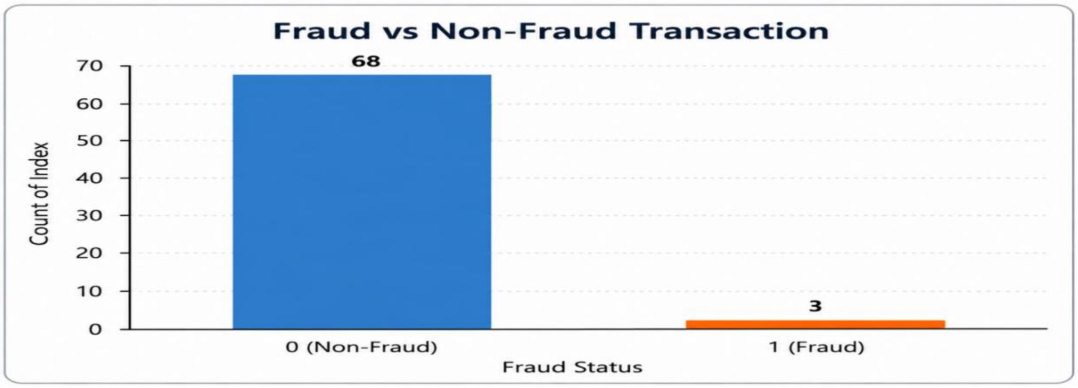

Figure 6.2: Fraud vs Non-Fraud Transaction Bar Chart

8. Fraud Count and Non-Fraud Count by Time

An Area Chart was used to visualize fraud and non-fraud transactions with respect to time. This chart helps in understanding when fraudulent transactions occurred and how transaction activities are distributed over different time periods.

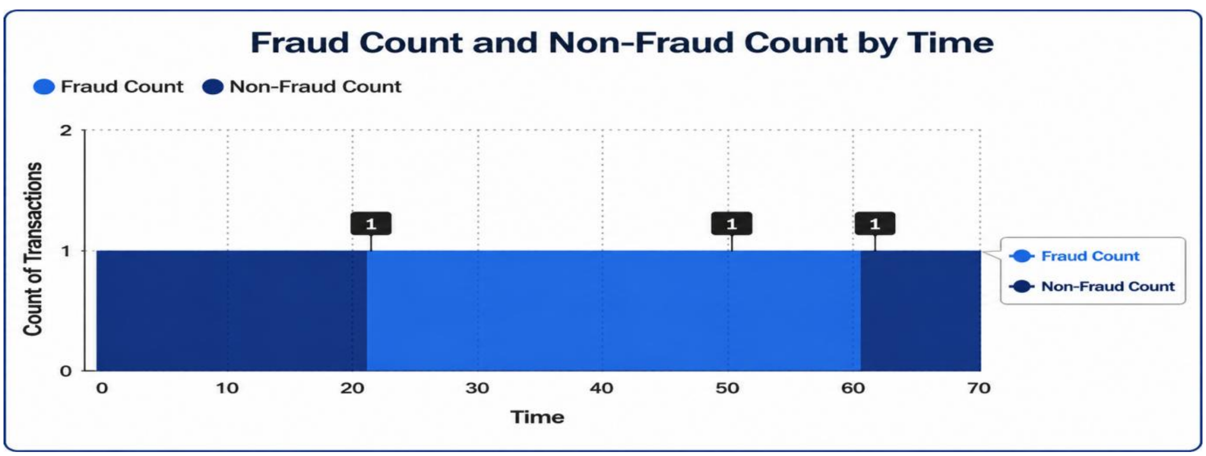

Figure 6.3: Fraud Count and Non-Fraud Count by Time

9. Fraud Distribution Donut Chart

A Donut Chart was created to display the percentage distribution of fraud and non-fraud transactions. The chart visually represents the share of fraudulent transactions compared to legitimate transactions.

The dashboard analysis shows that approximately 95.77% of transactions are normal, while only 4.23% are fraudulent.

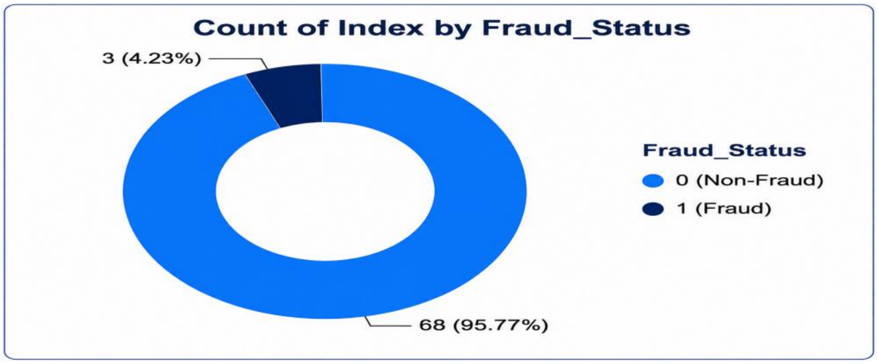

Figure 6.4: Fraud Distribution Donut Chart

10. Total Transactions by Time

A Line Chart was used to visualize the total number of transactions across different time intervals. The chart helps in understanding transaction activity and identifying patterns over time.

The visualization shows a mostly stable transaction trend, with a noticeable peak at **Time = 58**, indicating a higher transaction count during that interval. This helps users analyze transaction behavior and identify variations in transaction activity.

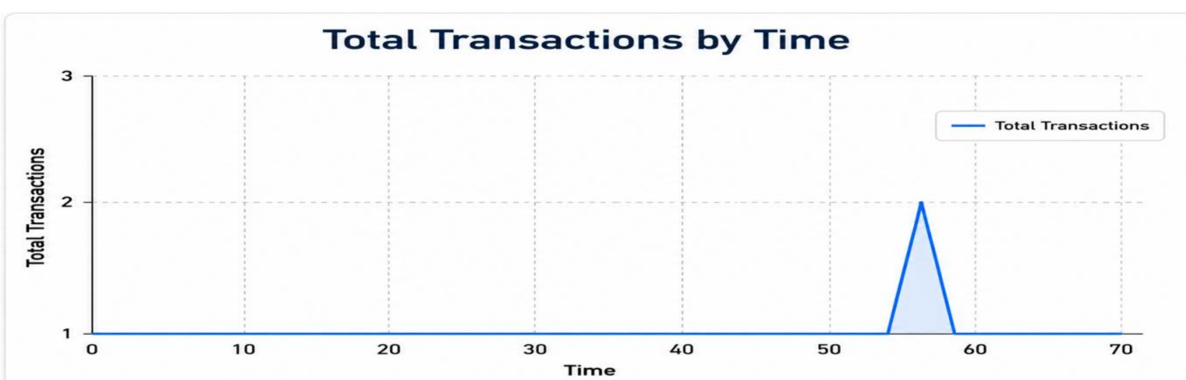

Figure 6.5: Total Transactions by Time.

❖ Final Dashboard

All visualizations were combined into a single interactive dashboard titled **"Credit Card Fraud Detection"**. The dashboard provides a comprehensive overview of transaction data and enables users to monitor fraud-related metrics efficiently.

The use of KPI cards, charts, and tables improves data interpretation and helps users gain meaningful insights from transaction records. The dashboard successfully transforms raw data into visual information that supports fraud analysis and reporting.

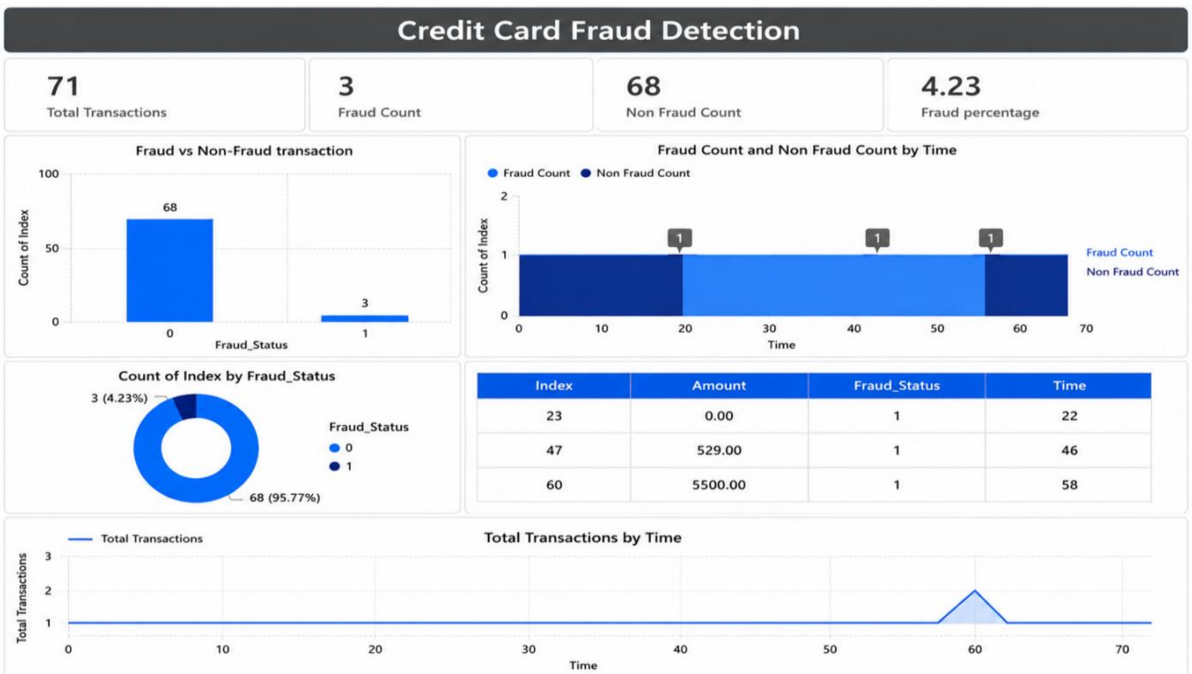

Figure 6.6: Final Credit Card Fraud Detection Dashboard

7.RESULT

The Credit Card Fraud Detection Dashboard was successfully developed using Microsoft Power BI. After performing data cleaning, transformation, and analysis, the processed data was visualized through interactive charts, KPI cards, and tables. The dashboard provides a clear overview of transaction activities and enables users to analyze fraudulent and non-fraudulent transactions effectively.

The analysis revealed that the dataset contains a total of **71 transactions**, out of which **3 transactions were identified as fraudulent** and **68 transactions were identified as non-fraudulent**. The calculated fraud percentage was **4.23%**, indicating that fraudulent transactions represent only a small portion of the overall transaction volume.

Dashboard Results

- Total Transactions: **71**
- Fraud Count: **3**
- Non-Fraud Count: **68**
- Fraud Percentage: **4.23%**

Discussion of Visualizations

KPI Cards

The KPI cards provide a quick summary of the dataset by displaying the total number of transactions, fraud count, non-fraud count, and fraud percentage. These indicators help users understand the overall fraud situation at a glance.

Fraud vs Non-Fraud Transaction Chart

The bar chart clearly shows that non-fraudulent transactions significantly outnumber fraudulent transactions. This indicates that fraud cases are relatively rare but still require attention due to their financial impact.

Fraud Count and Non-Fraud Count by Time

The chart displays the occurrence of fraudulent transactions at different time intervals. It helps identify when fraud transactions occurred and provides insight into transaction patterns over time.

Fraud Distribution Donut Chart

The donut chart shows that **95.77%** of transactions are normal, while only **4.23%** are fraudulent. This visualization effectively highlights the distribution of transaction categories within the dataset.

Fraud Transaction Details Table

The table visual displays detailed information about fraudulent transactions, including transaction amount, transaction time, and fraud status. This allows users to inspect suspicious records individually and verify fraud occurrences.

Total Transactions by Time

The line chart shows transaction activity across different time intervals. A noticeable peak is observed at **Time = 58**, where transaction activity increases compared to other intervals. This helps identify unusual transaction behavior and supports further investigation.

Overall Discussion

The dashboard successfully transforms raw transaction data into meaningful visual insights. Through the use of KPI cards, charts, and tables, users can easily understand transaction patterns and identify fraudulent activities. The results demonstrate the effectiveness of Power BI as a business intelligence tool for fraud analysis and reporting.

The developed dashboard provides a simple and interactive solution for monitoring credit card transactions and supports better decision-making by presenting fraud-related information in a clear and understandable format.

8. CONCLUSION

The Credit Card Fraud Detection Dashboard was successfully developed using Microsoft Power BI to analyze and visualize credit card transaction data. The project involved collecting the dataset, performing data cleaning and transformation using Power Query, generating analytical measures, and creating interactive visualizations. These processes helped convert raw transaction data into meaningful and easy-to-understand insights.

The dashboard effectively displays important metrics such as Total Transactions, Fraud Count, Non-Fraud Count, and Fraud Percentage through KPI cards, charts, and tables. The visualizations enabled clear comparison between fraudulent and non-fraudulent transactions and provided a better understanding of transaction patterns. The analysis revealed that only a small percentage of transactions were fraudulent, highlighting the importance of monitoring and detecting suspicious activities.

The project demonstrates the usefulness of Business Intelligence and Data Visualization techniques in fraud analysis. By using Power BI, complex transaction data can be presented in an interactive and user-friendly format, supporting efficient monitoring and decision-making. Overall, the developed dashboard provides a simple, effective, and practical solution for analyzing credit card fraud data and can serve as a foundation for future fraud detection and analytical systems.

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